

AWS Architecture & Database Services

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AWS Well-Architected Framework - Design Principles

1. Operational Excellence
2. Security
3. Reliability
4. Performance Efficiency
5. Cost Optimization
6. Sustainability

MySQL & PostgreSQL on RDS

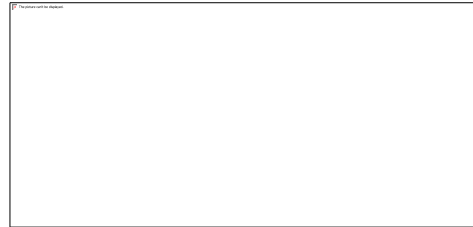
MySQL:

- Open-source, widely used
- High availability & read replicas

PostgreSQL:

- Advanced open-source database
- JSON support, extensibility
- Strong community support

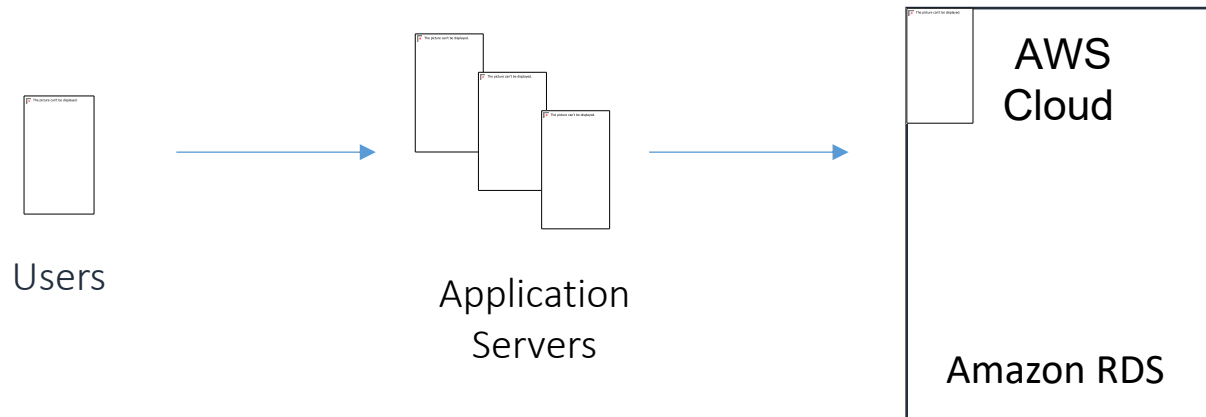
Relational Database Service (RDS)



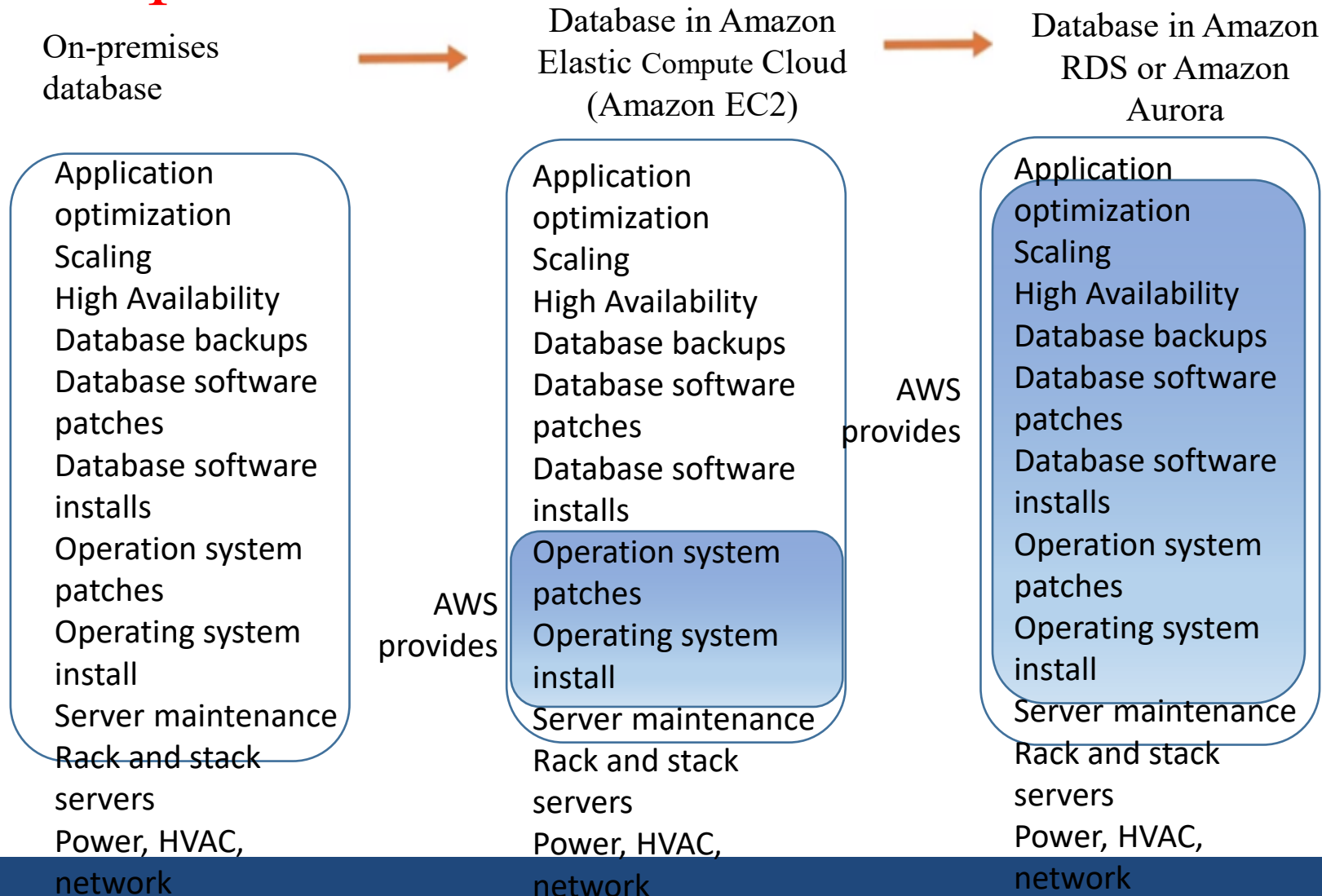
- Managed relational database service
- Automated backups, patching, and scaling
- Supports multiple engines: MySQL, PostgreSQL, SQL Server, Oracle, MariaDB
- High availability with Multi-AZ deployments

Amazon RDS

Managed service that sets up and operates a relational database in the cloud.



On-premises Databases to Amazon RDS



Amazon RDS DB instances

Amazon
RDS



Amazon RDS
DB main
instance

DB Instance Class

- CPU
- Memory
- Network performance

DB Instance Storage

- Magnetic
- General Purpose (solid state drive, or SSD)
- Provisioned IOPS

MySQL

Amazon Aurora

Microsoft SQL Server

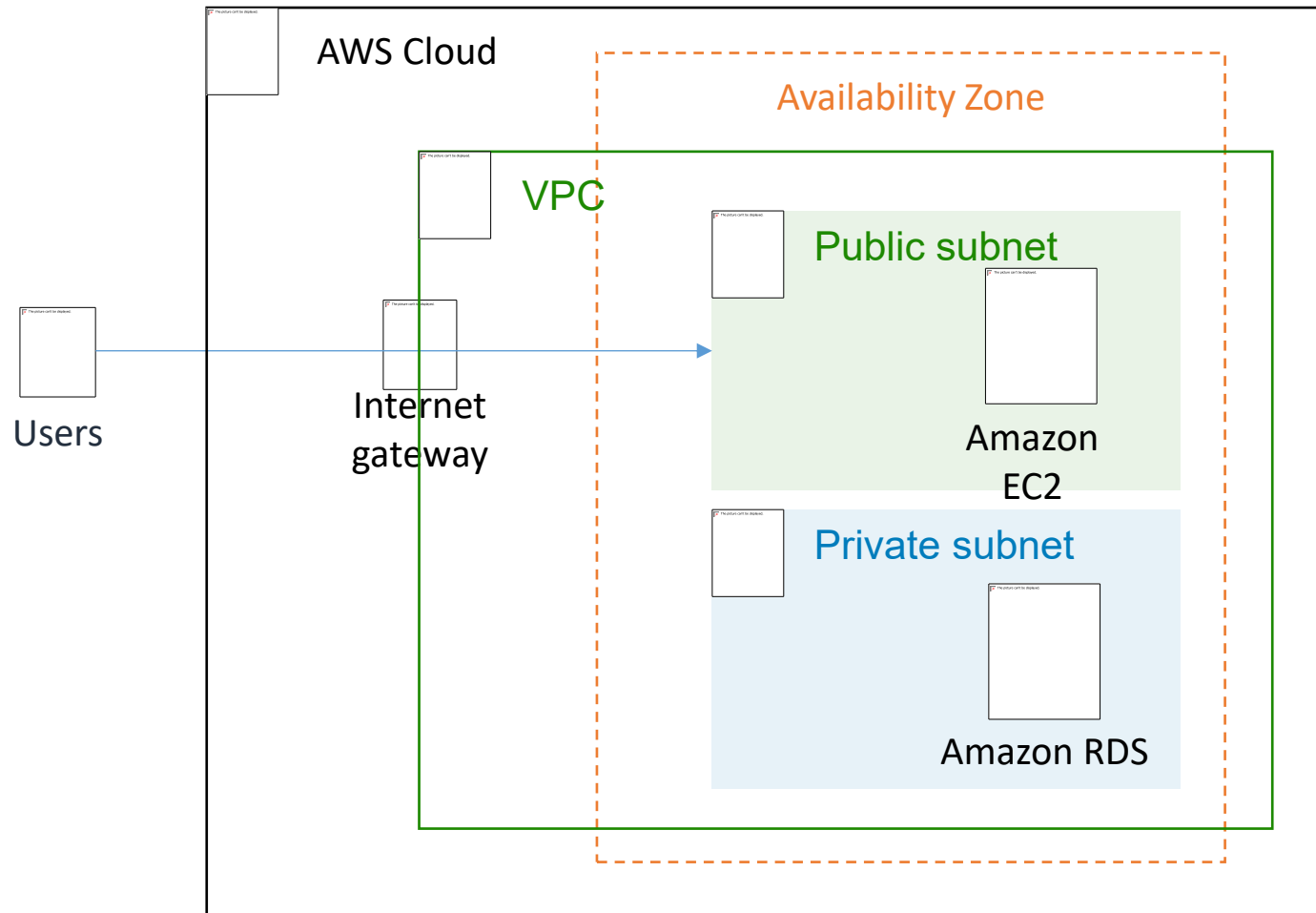
PostgreSQL

MariaDB

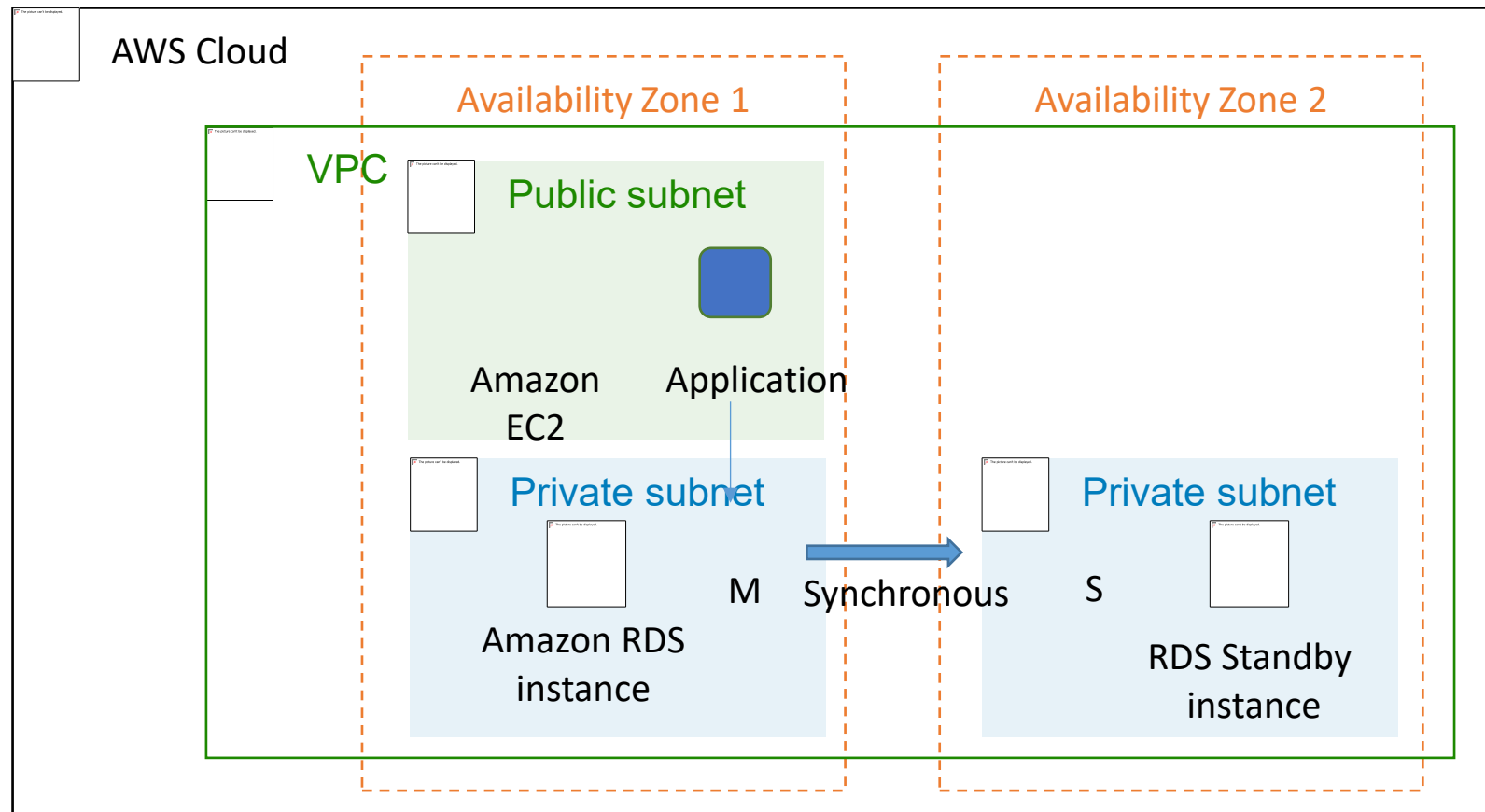
Oracle

DB engines

Amazon RDS in a Virtual Private Cloud (VPC)



High availability with Multi-AZ deployment



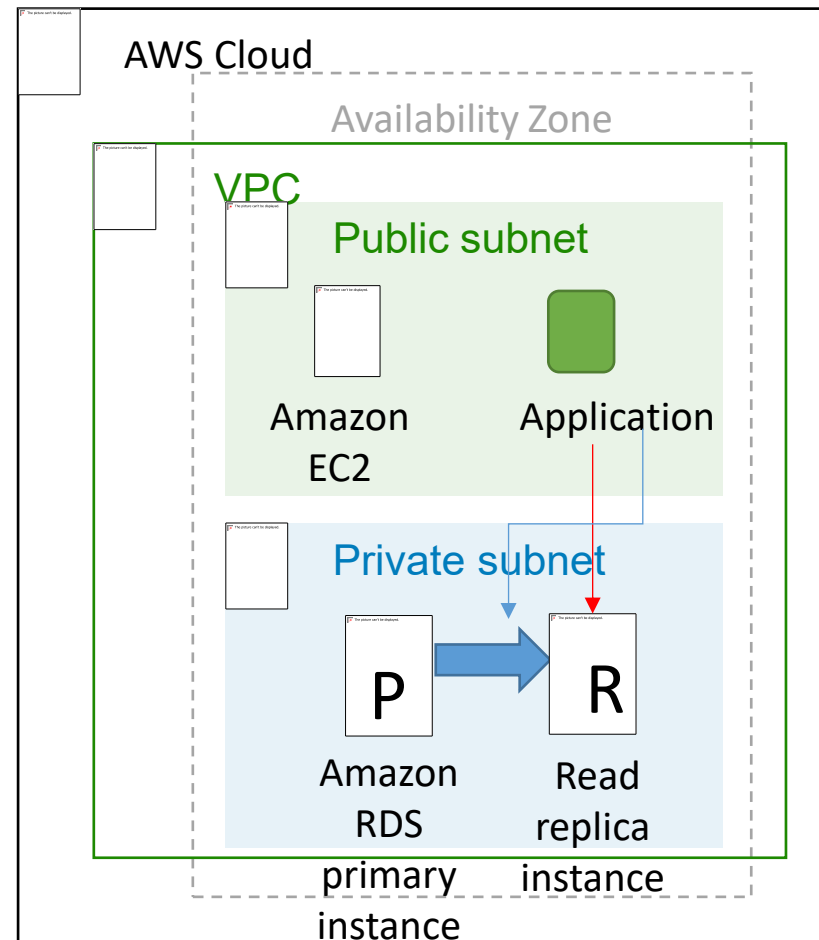
Amazon RDS read replicas

Features

- Offers asynchronous replication
- Can be promoted to primary if needed

Functionality

- Use for read-heavy database workloads
- Offload read queries



When to Use Amazon RDS

Use Amazon RDS when your application requires:

- Complex transactions or complex queries
- A medium to high query or write rate – Up to 30,000 IOPS (15,000 reads + 15,000 writes)
- No more than a single worker node or shard
- High durability

Do not use Amazon RDS when your application requires:

- Massive read/write rates (for example, 150,000 write/second)
- Sharding due to high data size or throughput demands
- Simple GET or PUT requests and queries that a NoSQL database can handle
- Relational database management system (RDBMS) customization

Amazon RDS: Deployment type and data transfer

Requests –

- The number of input and output requests that are made to the database

Deployment type—Storage and I/O charges vary, depending on whether you deploy to –

- Single Availability Zone
- Multiple Availability Zones

Data transfer –

- No charge for inbound data transfer
- Tiered charges for outbound data transfer

SQL Server on RDS

- Microsoft SQL Server managed on RDS
- Easy provisioning & scaling
- Supports Multi-AZ and backups
- Integration with Microsoft ecosystem

Amazon DynamoDB

Relational versus non-relational databases

	Relational (SQL)	Non-Relational								
Data Storage	Rows and columns	Key-value, document, graph								
Schemas	Fixed	Dynamic								
Querying	Uses SQL	Focuses on collection of documents								
Scalability	Vertical	Horizontal								
Example	<table><tr><th>ISBN</th><th>Title</th><th>Author</th><th>Format</th></tr><tr><td>3111111223439</td><td>Withering Depths</td><td>Jackson, Mateo</td><td>Paperback</td></tr></table>	ISBN	Title	Author	Format	3111111223439	Withering Depths	Jackson, Mateo	Paperback	<pre>{ ISBN: 3111111223439, Title: "Withering Depths", Author: "Jackson, Mateo", Format: "Paperback" }</pre>
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What is Amazon DynamoDB?

Fast and flexible NoSQL database service for any scale



Amazon DynamoDB

- NoSQL database tables
- Virtually unlimited storage
- Items can have differing attributes
- Low-latency queries
- Scalable read/write throughput

Amazon DynamoDB core components

- Tables, items, and attributes are the core DynamoDB components
 - A table is a collection of data.
 - Items are a group of attributes that is uniquely identifiable among all the other items.
 - Attributes are a fundamental data element, something that does not need to be broken down any further.
- DynamoDB supports two different kinds of primary keys:
Partition key and partition and sort key

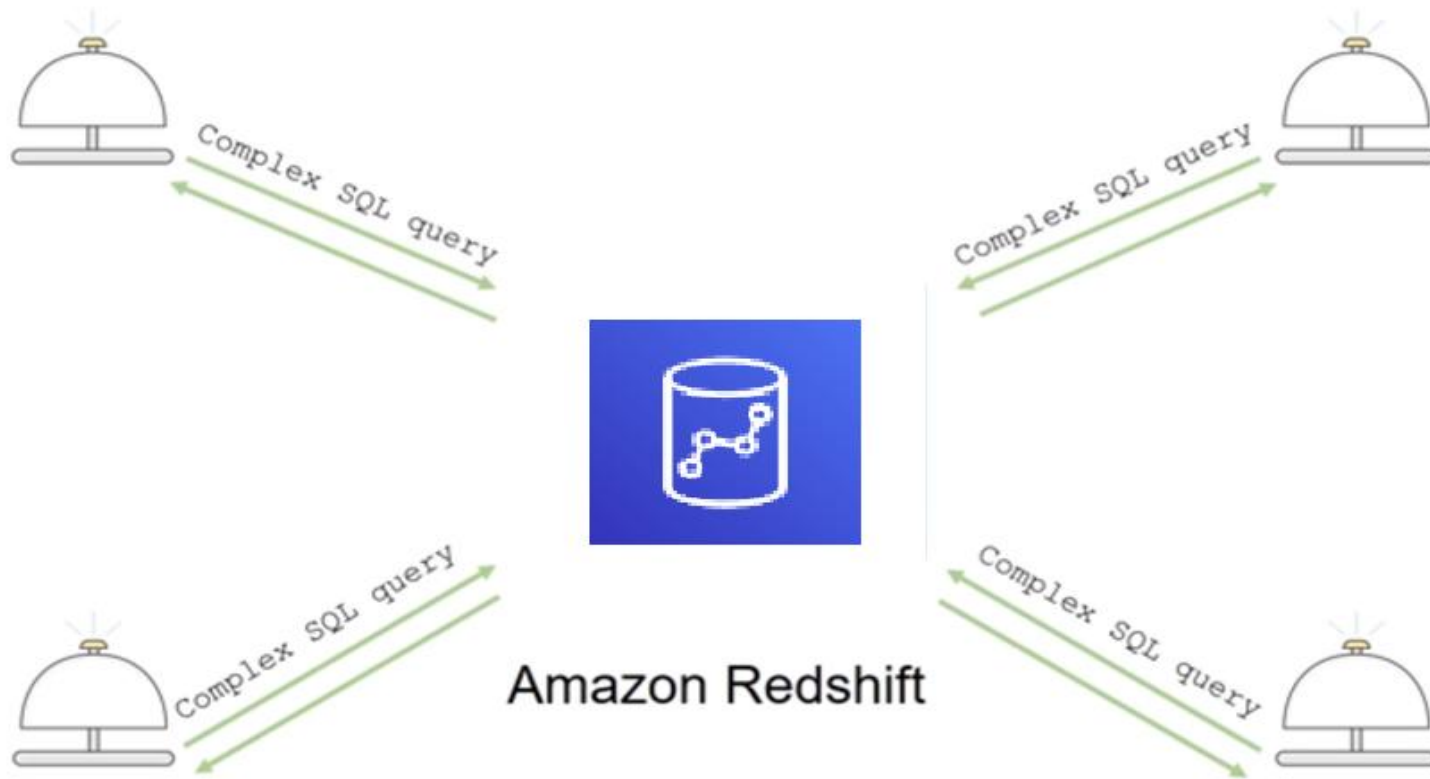
Amazon Redshift

Amazon Redshift

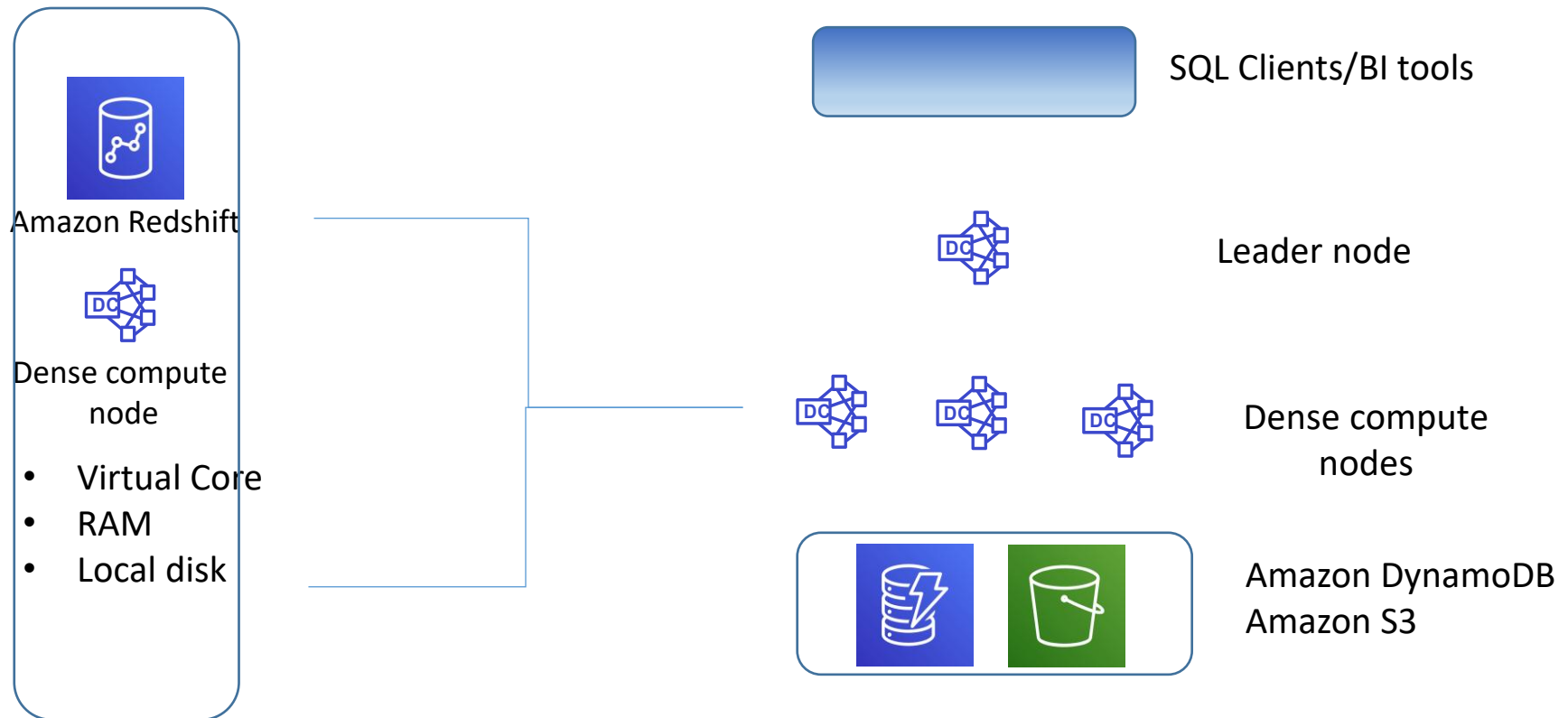


- Managed data warehouse service
- Columnar storage and parallel query execution
- Integrates with BI tools
- Scales to petabytes of data

Introduction to Amazon Redshift



Parallel processing architecture



Amazon Aurora

Amazon Aurora



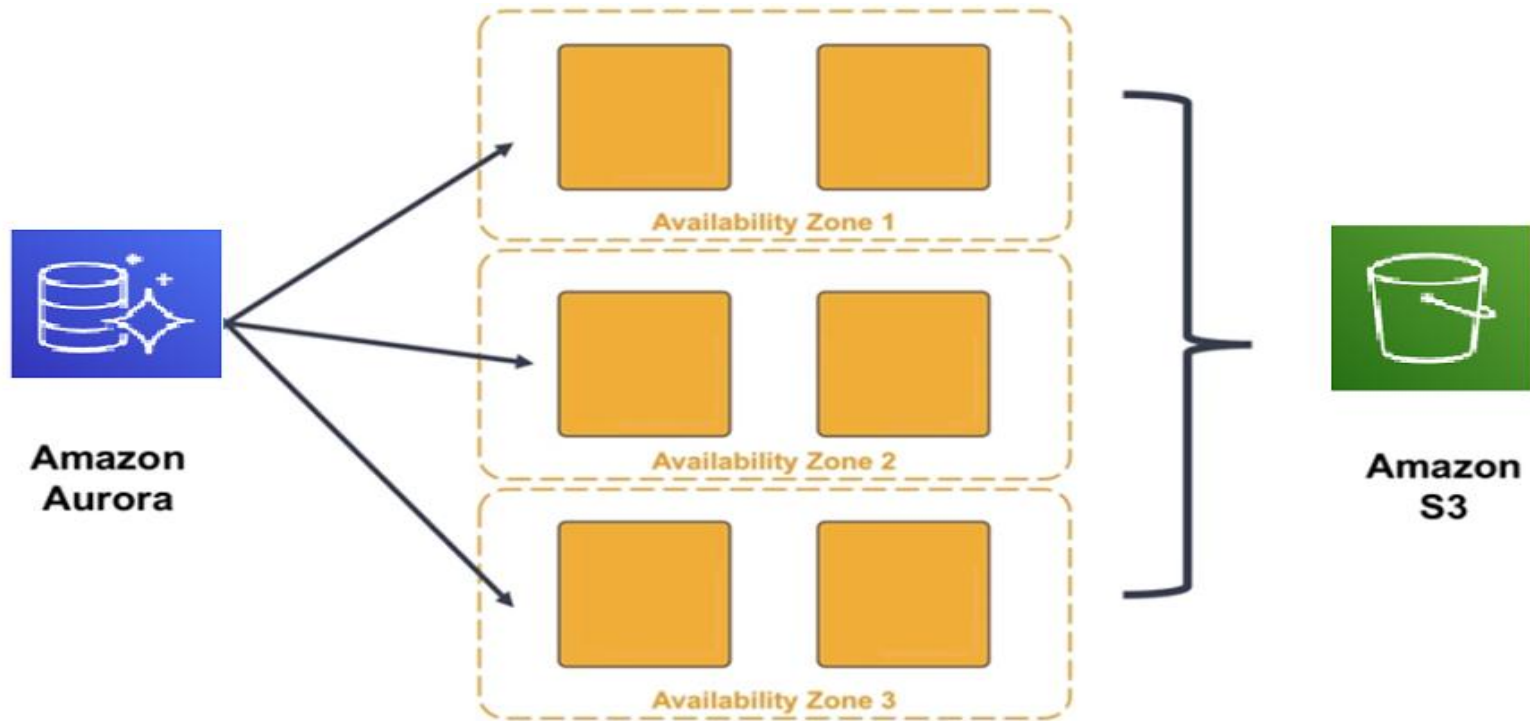
Amazon Aurora

- Enterprise-class relational database
- Compatible with MySQL or PostgreSQL
- Automate time-consuming tasks (such as provisioning, patching, backup, recovery, failure detection, and repair).

Amazon Aurora service benefits



High availability



Content

5.1 AWS Well-Architected Framework Design Principles

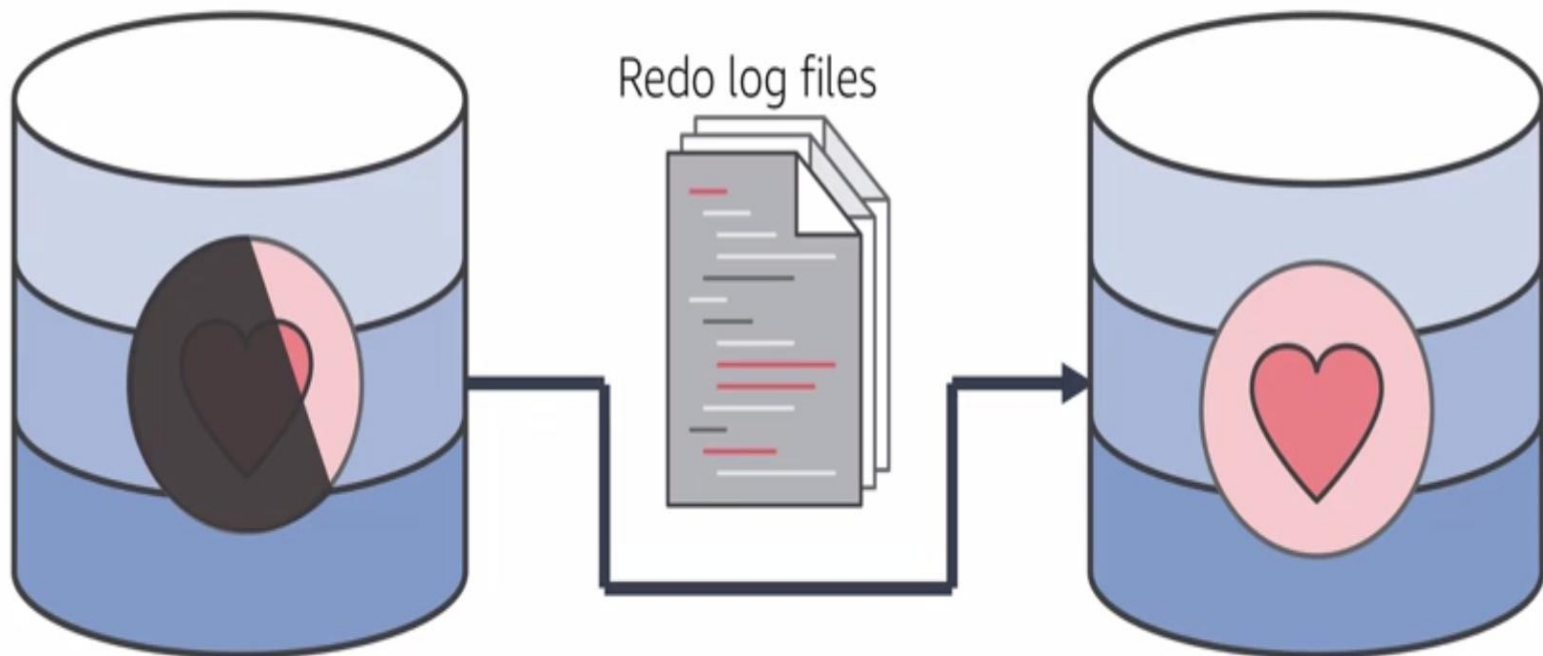
5.2 Relational Database Service (RDS)

5.3 Amazon DynamoDB

5.4 Amazon Redshift

5.5. Amazon Aurora

Resilient design



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